



《Linear Model》
Chapter 3
Simple Linear Regression
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Chapter 3 Simple Linear Regression

- **The Model**
- **Estimation of β_0 , β_1 , and σ^2**
- **Hypothesis Test and Confidence Interval for β_1**
- **Coefficient of Determination**

3.1 THE MODEL



By (1.1), the *simple linear regression* model for n observations can be written as

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, 2, \dots, n. \quad (6.1)$$

The designation *simple* indicates that there is only one x to predict the response y , and *linear* means that the model (6.1) is linear in β_0 and β_1 . [Actually, it is the assumption $E(y_i) = \beta_0 + \beta_1 x_i$ that is linear; see assumption 1 below.] For example, a model such as $y_i = \beta_0 + \beta_1 x_i^2 + \varepsilon_i$ is linear in β_0 and β_1 , whereas the model $y_i = \beta_0 + e^{\beta_1 x_i} + \varepsilon_i$ is not linear.

3.1 THE MODEL



To complete the model in (6.1), we make the following additional assumptions:

1. $E(\varepsilon_i) = 0$ for all $i = 1, 2, \dots, n$, or, equivalently, $E(y_i) = \beta_0 + \beta_1 x_i$.
2. $\text{var}(\varepsilon_i) = \sigma^2$ for all $i = 1, 2, \dots, n$, or, equivalently, $\text{var}(y_i) = \sigma^2$.
3. $\text{cov}(\varepsilon_i, \varepsilon_j) = 0$ for all $i \neq j$, or, equivalently, $\text{cov}(y_i, y_j) = 0$.

Assumption 1 states that the model (6.1) is correct, implying that y_i depends only on x_i and that all other variation in y_i is random. Assumption 2 asserts that the variance of ε or y does not depend on the values of x_i . (Assumption 2 is also known as the assumption of *homoscedasticity*, *homogeneous variance* or *constant variance*.) Under assumption 3, the ε variables (or the y variables) are uncorrelated with each other. In Section 6.3, we will add a normality assumption, and the y (or the ε) variables will thereby be independent as well as uncorrelated. Each assumption has been stated in terms of the ε 's or the y 's. For example, if $\text{var}(\varepsilon_i) = \sigma^2$, then $\text{var}(y_i) = E[y_i - E(y_i)]^2 = E(y_i - \beta_0 - \beta_1 x_i)^2 = E(\varepsilon_i^2) = \sigma^2$.

3.2 ESTIMATION OF β_0 , β_1 , AND σ^2

Using a random sample of n observations $y_1, y_2, \dots, y_n$ and the accompanying fixed values $x_1, x_2, \dots, x_n$, we can estimate the parameters β_0 , β_1 , and σ^2 . To obtain the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$, we use the method of least squares, which does not require any distributional assumptions (for maximum likelihood estimators based on normality, see Section 7.6.2).

In the *least-squares* approach, we seek estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize the sum of squares of the deviations $y_i - \hat{y}_i$ of the n observed y_i 's from their predicted values $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$:

$$\hat{\boldsymbol{\varepsilon}}' \hat{\boldsymbol{\varepsilon}} = \sum_{i=1}^n \hat{\varepsilon}_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2. \quad (6.2)$$

3.2 ESTIMATION OF β_0 , β_1 , AND σ^2

Note that the predicted value $\hat{y}_i$ estimates $E(y_i)$, not y_i ; that is, $\hat{\beta}_0 + \hat{\beta}_1 x_i$ estimates $\beta_0 + \beta_1 x_i$, not $\beta_0 + \beta_1 x_i + \varepsilon_i$. A better notation would be $\widehat{E}(y_i)$, but $\hat{y}_i$ is commonly used.

To find the values of $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize $\hat{\boldsymbol{\varepsilon}}' \hat{\boldsymbol{\varepsilon}}$ in (6.2), we differentiate with respect to $\hat{\beta}_0$ and $\hat{\beta}_1$ and set the results equal to 0:

$$\frac{\partial \hat{\boldsymbol{\varepsilon}}' \hat{\boldsymbol{\varepsilon}}}{\partial \hat{\beta}_0} = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0, \quad (6.3)$$

$$\frac{\partial \hat{\boldsymbol{\varepsilon}}' \hat{\boldsymbol{\varepsilon}}}{\partial \hat{\beta}_1} = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) x_i = 0. \quad (6.4)$$

3.2 ESTIMATION OF β_0 , β_1 , AND σ^2

The solution to (6.3) and (6.4) is given by

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}} \quad (6.5)$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}. \quad (6.6)$$

To verify that $\hat{\beta}_0$ and $\hat{\beta}_1$ in (6.5) and (6.6) minimize $\hat{\epsilon}'\hat{\epsilon}$ in (6.2), we can examine the second derivatives or simply observe that $\hat{\epsilon}'\hat{\epsilon}$ has no maximum and therefore the first derivatives yield a minimum. For an algebraic proof that $\hat{\beta}_0$ and $\hat{\beta}_1$ minimize (6.2), see (7.10) in Section 7.3.1.

3.2 ESTIMATION OF β_0 , β_1 , AND σ^2

Example 6.2. Students in a statistics class (taught by one of the authors) claimed that doing the homework had not helped prepare them for the midterm exam. The exam score y and homework score x (averaged up to the time of the midterm) for the 18 students in the class were as follows:

y	x	y	x	y	x
95	96	72	89	35	0
80	77	66	47	50	30
0	0	98	90	72	59
0	0	90	93	55	77
79	78	0	18	75	74
77	64	95	86	66	67

3.2 ESTIMATION OF β_0 , β_1 , AND σ^2

Using (6.5) and (6.6), we obtain

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2} \\ &= \frac{81,195 - 18(58.056)(61.389)}{80,199 - 18(58.056)^2} = .8726, \\ \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} = 61.389 - .8726(58.056) = 10.73.\end{aligned}$$

The prediction equation is thus given by

$$\hat{y} = 10.73 + .8726x.$$

This equation and the 18 points are plotted in Figure 6.1. It is readily apparent in the plot that the slope $\hat{\beta}_1$ is the rate of change of $\hat{y}$ as x varies and that the intercept $\hat{\beta}_0$ is the value of $\hat{y}$ at $x = 0$.

3.2 ESTIMATION OF β_0 , β_1 , AND σ^2

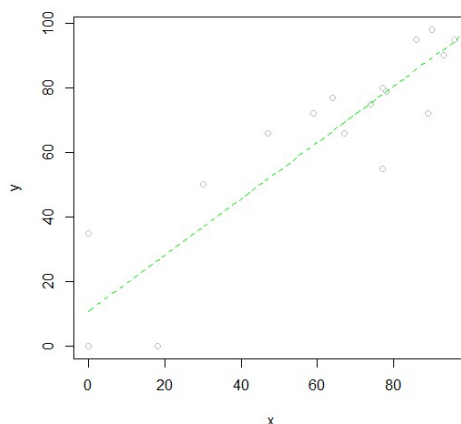
```
# The exam score y and homework score x
x<- c(96, 89, 0, 77, 47, 30, 0, 90, 59, 0, 93, 77, 78, 18, 74, 64, 86, 67)
y<-c(95, 72, 35, 80, 66, 50, 0, 98, 72, 0, 90, 55, 79, 0, 75, 77, 95, 66)
beta1=(x%*%y-length(x)*mean(x)*mean(y))/(x%*%x-length(x)*mean(x)*mean(x))
beta0=mean(y)-beta1*mean(x)

x1=x-mean(x)
y1=y-mean(y)
betanew1=(x1%*%y1)/(x1%*%x1)
betanew0=mean(y)-betanew1*mean(x)

lm1<-lm(y~x)
summary(lm1)

plot(x,y,col="gray")

x3=c(0:100)
y3=beta0+beta1*x3
lines(x3,y3,lty=2,col="green")
```



3.2 ESTIMATION OF β_0 , β_1 , AND σ^2

assumptions, we obtain the following means and variances of $\hat{\beta}_0$ and $\hat{\beta}_1$:

$$E(\hat{\beta}_1) = \beta_1 \quad (6.7)$$

$$E(\hat{\beta}_0) = \beta_0 \quad (6.8)$$

$$\text{var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (6.9)$$

$$\text{var}(\hat{\beta}_0) = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]. \quad (6.10)$$

Note that in discussing $E(\hat{\beta}_1)$ and $\text{var}(\hat{\beta}_1)$, for example, we are considering random variation of $\hat{\beta}_1$ from sample to sample. It is assumed that the n values $x_1, x_2, \dots, x_n$ would remain the same in future samples so that $\text{var}(\hat{\beta}_1)$ and $\text{var}(\hat{\beta}_0)$ are constant.

3.2 ESTIMATION OF β_0 , β_1 , AND σ^2

In (6.9), we see that $\text{var}(\hat{\beta}_1)$ is minimized when $\sum_{i=1}^n (x_i - \bar{x})^2$ is maximized. If the x_i values have the range $a \leq x_i \leq b$, then $\sum_{i=1}^n (x_i - \bar{x})^2$ is maximized if half the x 's are selected equal to a and half equal to b (assuming that n is even; see Problem 6.4). In (6.10), it is clear that $\text{var}(\hat{\beta}_0)$ is minimized when $\bar{x} = 0$.

The method of least squares does not yield an estimator of $\text{var}(y_i) = \sigma^2$; minimization of $\hat{\epsilon}'\hat{\epsilon}$ yields only $\hat{\beta}_0$ and $\hat{\beta}_1$. To estimate σ^2 , we use the definition in (3.6), $\sigma^2 = E[y_i - E(y_i)]^2$. By assumption 2 in Section 6.1, σ^2 is the same for each y_i , $i = 1, 2, \dots, n$. Using $\hat{y}_i$ as an estimator of $E(y_i)$, we estimate σ^2 by an average from the sample, that is

$$s^2 = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - 2} = \frac{\sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2}{n - 2} = \frac{\text{SSE}}{n - 2}, \quad (6.11)$$

where $\hat{\beta}_0$ and $\hat{\beta}_1$ are given by (6.5) and (6.6) and $\text{SSE} = \sum_i (y_i - \hat{y}_i)^2$. The deviation $\hat{\epsilon}_i = y_i - \hat{y}_i$ is often called the *residual* of y_i , and SSE is called the *residual sum of squares* or *error sum of squares*. With $n - 2$ in the denominator, s^2 is an unbiased

3.2 ESTIMATION OF β_0 , β_1 , AND σ^2

estimator of σ^2 :

$$E(s^2) = \frac{E(\text{SSE})}{n-2} = \frac{(n-2)\sigma^2}{n-2} = \sigma^2. \quad (6.12)$$

Intuitively, we divide by $n-2$ in (6.11) instead of $n-1$ as in $s^2 = \sum_i (y_i - \bar{y})^2 / (n-1)$ in (5.6), because $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ has two estimated parameters and should thereby be a better estimator of $E(y_i)$ than $\bar{y}$. Thus we expect $\text{SSE} = \sum_i (y_i - \hat{y}_i)^2$ to be less than $\sum_i (y_i - \bar{y})^2$. In fact, using (6.5) and (6.6), we can write the numerator of (6.11) in the form

$$\text{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \bar{y})^2 - \frac{\left[\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \right]^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad (6.13)$$

$\swarrow$ $\frac{S_{xy}^2}{S_{xx}}$
 $= SST - SSR$

which shows that $\sum_i (y_i - \hat{y}_i)^2$ is indeed smaller than $\sum_i (y_i - \bar{y})^2$.

3.3 HYPOTHESIS TEST AND CONFIDENCE INTERVAL FOR β_1

Typically, hypotheses about β_1 are of more interest than hypotheses about β_0 , since our first priority is to determine whether there is a linear relationship between y and x . (See Problem 6.9 for a test and confidence interval for β_0 .) In this section, we consider the hypothesis $H_0: \beta_1 = 0$, which states that there is no linear relationship between y and x in the model $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$. The hypothesis $H_0: \beta_1 = c$ (for $c \neq 0$) is of less interest.

In order to obtain a test for $H_0: \beta_1 = 0$, we assume that y_i is $N(\beta_0 + \beta_1 x_i, \sigma^2)$. Then $\hat{\beta}_1$ and s^2 have the following properties (these are special cases of results established in Theorem 7.6b in Section 7.6.3):

1. $\hat{\beta}_1$ is $N[\beta_1, \sigma^2 / \sum_i (x_i - \bar{x})^2]$.
2. $(n-2)s^2 / \sigma^2$ is $\chi^2(n-2)$.
3. $\hat{\beta}_1$ and s^2 are independent.

3.3 HYPOTHESIS TEST AND CONFIDENCE INTERVAL FOR β_1

From these three properties it follows by (5.29) that

$$H_0: \beta_1 = 0 \quad H_1: \beta_1 \neq 0$$

$$t = \frac{\hat{\beta}_1}{s / \sqrt{\sum_i (x_i - \bar{x})^2}} \quad (6.14)$$

is distributed as $t(n-2, \delta)$, the noncentral t with noncentrality parameter δ .

By a comment following (5.29), δ is given by $\delta = E(\hat{\beta}_1) / \sqrt{\text{var}(\hat{\beta}_1)} = \beta_1 / [\sigma / \sqrt{\sum_i (x_i - \bar{x})^2}]$. If $\beta_1 = 0$, then by (5.28), t is distributed as $t(n-2)$. For a two-sided alternative hypothesis $H_1: \beta_1 \neq 0$, we reject $H_0: \beta_1 = 0$ if $|t| \geq t_{\alpha/2, n-2}$, where $t_{\alpha/2, n-2}$ is the upper $\alpha/2$ percentage point of the central t distribution and α is the desired significance level of the test (probability of rejecting H_0 when it is true). Alternatively, we reject H_0 if $p \leq \alpha$, where p is the p value. For a two-sided test, the p value is defined as twice the probability that $t(n-2)$ exceeds the absolute value of the observed t .

3.3 HYPOTHESIS TEST AND CONFIDENCE INTERVAL FOR β_1

A $100(1 - \alpha)\%$ confidence interval for β_1 is given by

$$\hat{\beta}_1 \pm t_{\alpha/2, n-2} \frac{s}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}}. \quad (6.15)$$

Confidence intervals are defined and discussed further in Section 8.6. A confidence interval for $E(y)$ and a prediction interval for y are also given in Section 8.6.

3.3 HYPOTHESIS TEST AND CONFIDENCE INTERVAL FOR β_1

Example 6.3. We test the hypothesis $H_0: \beta_1 = 0$ for the grades data in Example 6.2. By (6.14), the t statistic is

$$t = \frac{\hat{\beta}_1}{s / \sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}} = \frac{.8726}{(13.8547) / (139.753)} = 8.8025.$$

Since $t = 8.8025 > t_{.025, 16} = 2.120$, we reject $H_0: \beta_1 = 0$ at the $\alpha = .05$ level of significance. Alternatively, the p value is 1.571×10^{-7} , which is less than .05.

A 95% confidence interval for β_1 is given by (6.15) as

$$\hat{\beta}_1 \pm t_{.025, 16} \frac{s}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

$$.8726 \pm 2.120(.09914)$$

$$.8726 \pm .2102$$

$$(.6624, 1.0828).$$

3.3 HYPOTHESIS TEST AND CONFIDENCE INTERVAL FOR β_1

```
x<- c(96, 89, 0, 77, 47, 30, 0, 90, 59, 0, 93, 77, 78, 18, 74, 64, 86, 67)
y<-c(95, 72, 35, 80, 66, 50, 0, 98, 72, 0, 90, 55, 79, 0, 75, 77, 95, 66)
lm1<-lm(y~x)
summary(lm1)
x1=x-mean(x)
s=13.85
beta1=0.87265
tfenweishu=qt(0.975,16)
#注意qt(0.025,16)=-qt(0.975,16)
b=tfenweishu*s/sqrt(x1%*%x1)
qujianbeta1=c(beta1-b,beta1+b)
```

```
Call:
lm(formula = y ~ x)

Residuals:
    Min       1Q   Median       3Q      Max
-26.4345  -8.8437   0.3528   9.6466  24.2731

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  10.72691     6.61733   1.621   0.125
x              0.87265     0.09914   8.802 1.57e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.85 on 16 degrees of freedom
Multiple R-squared:  0.8288,    Adjusted R-squared:  0.8181
F-statistic: 77.48 on 1 and 16 DF,  p-value: 1.571e-07
```

3.4 COEFFICIENT OF DETERMINATION

The *coefficient of determination* r^2 is defined as

$$r^2 = \frac{SSR}{SST} = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (6.16)$$

where $SSR = \sum_i (\hat{y}_i - \bar{y})^2$ is the regression sum of squares and $SST = \sum_i (y_i - \bar{y})^2$ is the total sum of squares. The total sum of squares can be partitioned into $SST = SSR + SSE$, that is,

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2. \quad (6.17)$$

$$SST = SSR + SSE$$

3.4 COEFFICIENT OF DETERMINATION

Thus r^2 in (6.16) gives the proportion of variation in y that is explained by the model or, equivalently, accounted for by regression on x .

We have labeled (6.16) as r^2 because it is the same as the square of the sample correlation coefficient r between y and x

$$r = \frac{s_{xy}}{\sqrt{s_x^2 s_y^2}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{[\sum_{i=1}^n (x_i - \bar{x})^2] [\sum_{i=1}^n (y_i - \bar{y})^2]}}, \quad (6.18)$$

where s_{xy} is given by 5.15 (see Problem 6.11). When x is a random variable, r estimates the population correlation in (3.19). The coefficient of determination r^2 is discussed further in Sections 7.7, 10.4, and 10.5.

3.4 COEFFICIENT OF DETERMINATION

Example 6.4. For the grades data of Example 6.2, we have

$$r^2 = \frac{SSR}{SST} = \frac{14,873.0}{17,944.3} = .8288.$$

The correlation between homework score and exam score is $r = \sqrt{.8288} = .910$.

The t statistic in (6.14) can be expressed in terms of r as follows:

$$t = \frac{\hat{\beta}_1}{s / \sqrt{\sum_i (x_i - \bar{x})^2}} \quad (6.19)$$

$$= \frac{\sqrt{n-2} r}{\sqrt{1-r^2}}. \quad (6.20)$$

3.4 COEFFICIENT OF DETERMINATION

```
> lml<-lm(y~x)
> summary(lml)
```

```
Call:
lm(formula = y ~ x)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-26.4345  -8.8437   0.3528   9.6466  24.2731
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  10.72691     6.61733   1.621   0.125
x             0.87265     0.09914   8.802 1.57e-07 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 13.85 on 16 degrees of freedom
Multiple R-squared:  0.8288,    Adjusted R-squared:  0.8181
F-statistic: 77.48 on 1 and 16 DF,  p-value: 1.571e-07
```

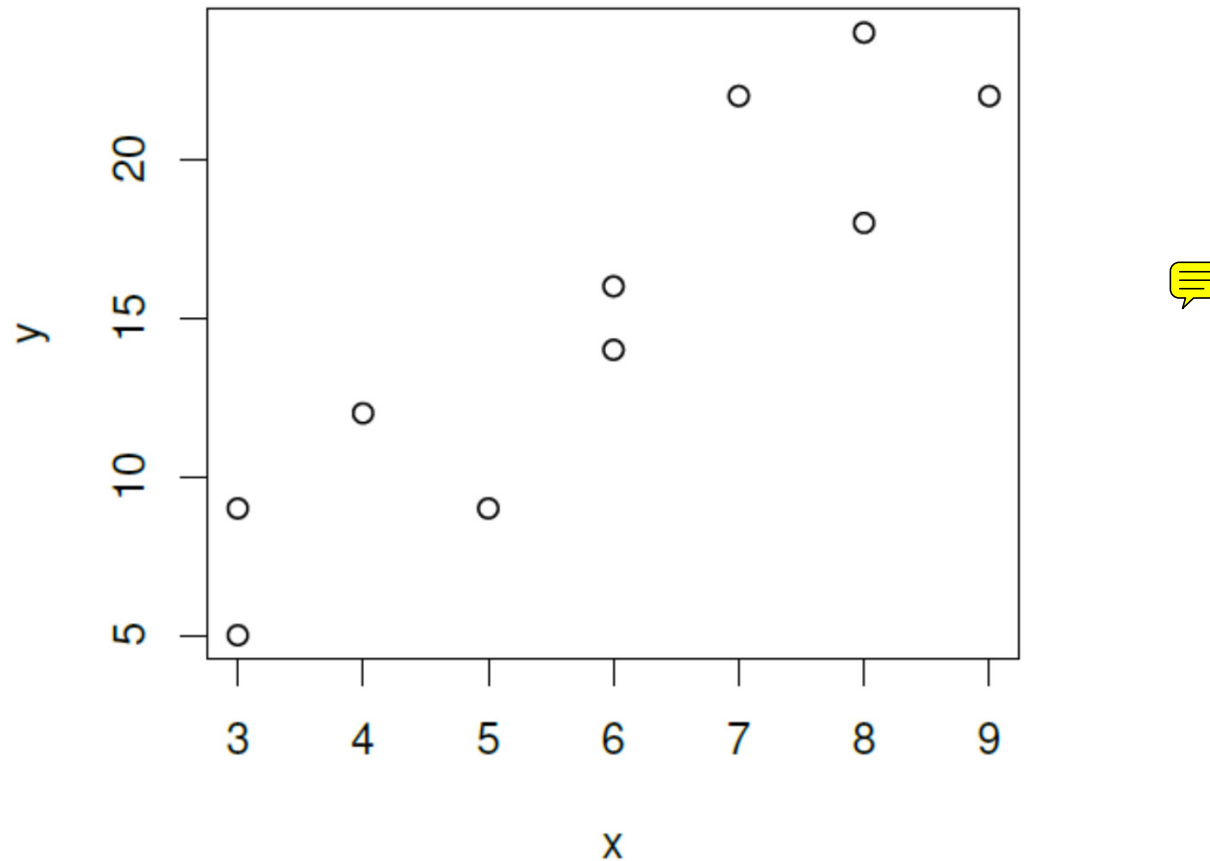
11.1 Introduction

Example 11.1.1 In one stage of the development of a new drug, an experiment is conducted to study how different dosages of the drug affect the duration of relief from the allergic symptoms. Ten patients are included in the experiment.

Dosage (x)	3	3	4	5	6	6	7	8	8	9
Duration of relief (y)	9	5	12	9	14	16	22	18	24	22

Example 11.1.1 (continued)

Dosage (x)	3	3	4	5	6	6	7	8	8	9
Duration of relief (y)	9	5	12	9	14	16	22	18	24	22



First step in the analysis: Drawing a scatterplot

Linear Regression

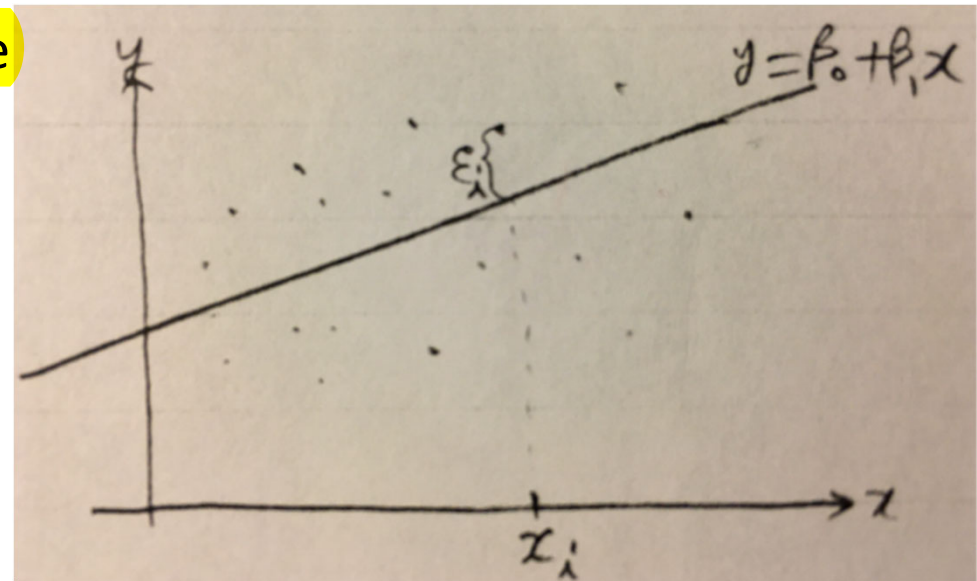
Objective

1. Find out whether a significant linear relationship exists between two variables
2. Use it to predict a future values

Notation

x : independent (predictor) variable

y : dependent (response) variable



x : 自变量

y : 响应变量, 因变量

Statistical model for a linear regression

$$y = \beta_0 + \beta_1 x + \varepsilon$$

ε : random error term

β_0 & β_1 : unknown

误差, - 假设 $\varepsilon_1, \dots, \varepsilon_n$ iid $\sim N(0, \sigma^2)$

回归线

x : 自变量, 非随机
用最小二乘法估计 β_0 和 β_1

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad 1 \leq i \leq n$$

Assumptions of linear regression

1. The relationship is linear and $E(\varepsilon_i) = 0$ for all i .
2. The errors have the same variance, i.e.,
 $\text{Var}(\varepsilon_i) = \sigma^2$ for all i .
3. $\varepsilon_i \sim N(0, \sigma^2)$ for all i . *independent*

Linear relationship

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

$\hat{\beta}_0$: intercept (value of y when $x = 0$)

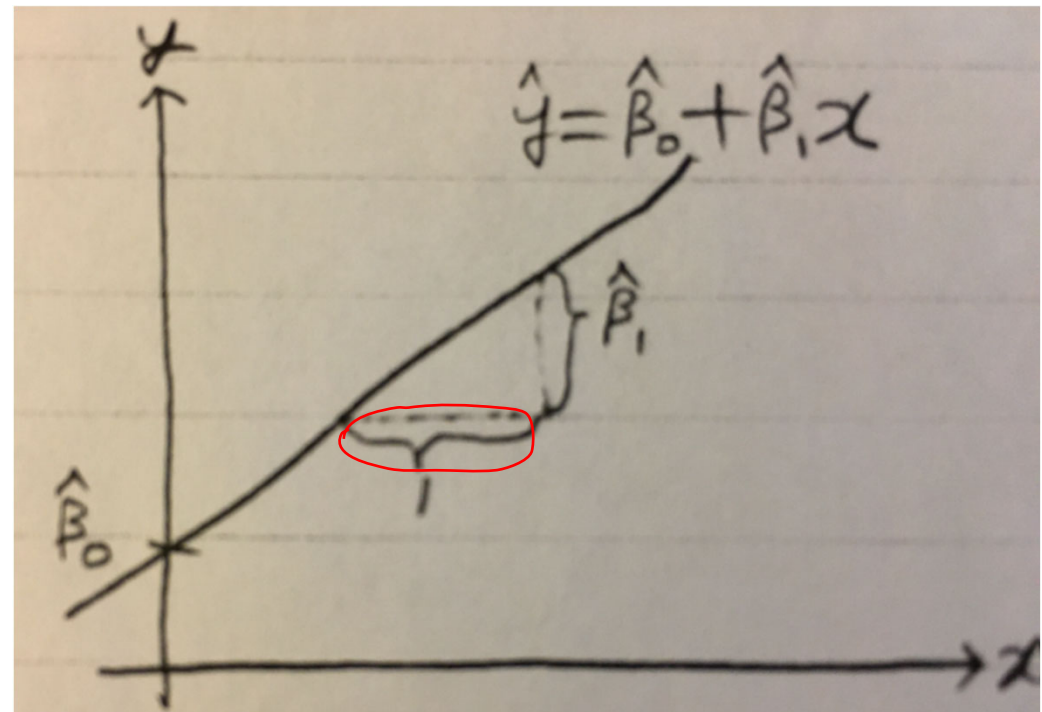
$\hat{\beta}_1$: slope (increase in y for each unit increase in x)

x 增加单位

对 y 值增加 $\hat{\beta}_1$

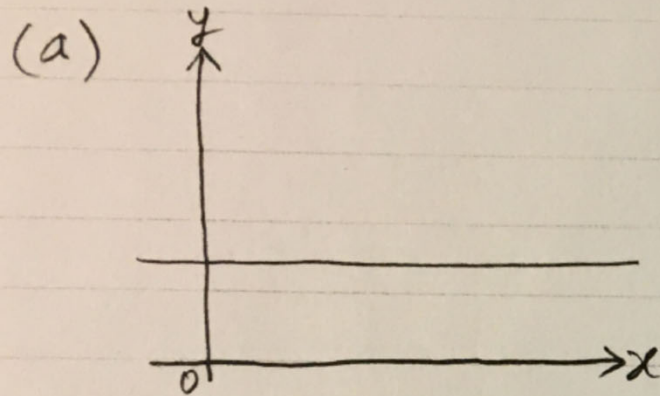
$$\hat{\beta}_0 =$$

$$\hat{\beta}_1 =$$

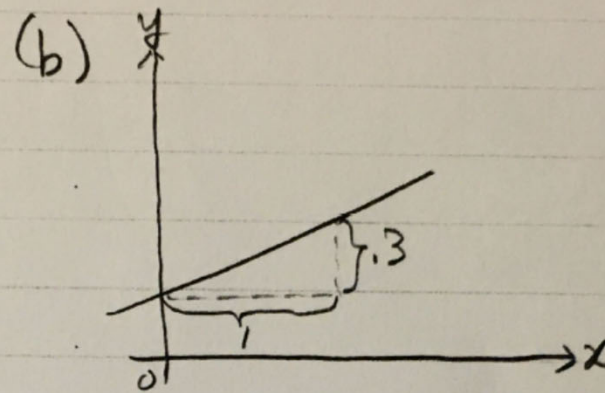


Some possibilities

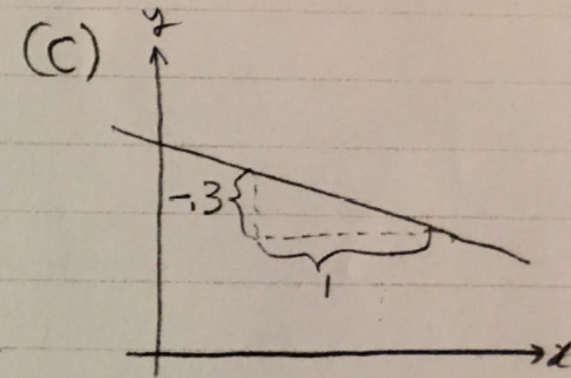
Some possibilities



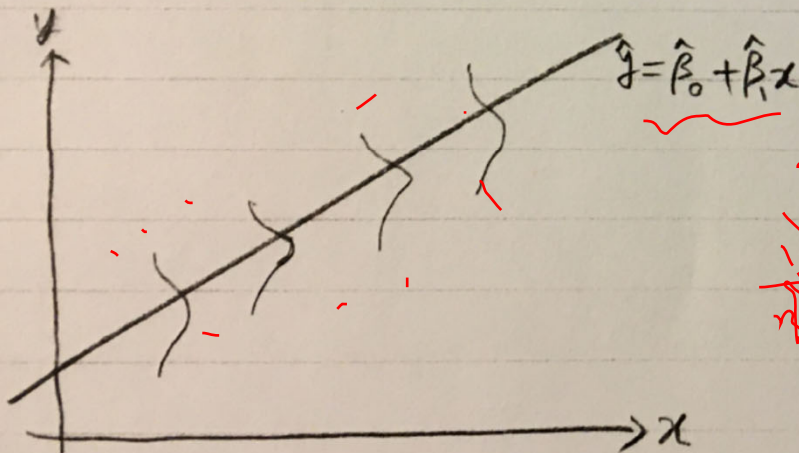
$\hat{\beta}_1 = 0$
No relationship
between x & y .



$\hat{\beta}_1 = 0.3$: positive
 y increases as
 x increases.



$\hat{\beta}_1 = -0.3$: negative
 y decreases as
 x increases.



经验回归直线
或经验回归方程

$\varepsilon_1, \dots, \varepsilon_n \text{ iid } \sim N(0, \sigma^2)$ β_0, β_1 未知参数
11.2 Least Squares Estimation 已知参数

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \Rightarrow \varepsilon_i = y_i - (\beta_0 + \beta_1 x_i), \quad i = 1, \dots, n$$

$$Y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$$

Likelihood function:

$$L = \prod_{i=1}^n (2\pi\sigma^2)^{-1/2} \exp \left[-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2} \right]$$

$$= (2\pi\sigma^2)^{-n/2} \exp \left[-\frac{\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2} \right]$$

Log likelihood:

$$l = \ln L = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}$$

Least-squares method: determination of β_0 and β_1 to minimize

$$\min \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 = \sum_{i=1}^n \varepsilon_i^2 \quad (\sum_{i=1}^n \varepsilon_i = 0)$$

The best fitting straight line is given by

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

$$E y_i = \beta_0 + \beta_1 x_i + 0$$

$$\text{Var } y_i = \text{Var } \varepsilon_i = \sigma^2$$

$$E \varepsilon_i = 0$$

估计值 $\hat{\beta}_0, \hat{\beta}_1$

拟合直线, 回归方程

Least-squares equation

求导

$$l = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}$$

$$\frac{\partial l}{\partial \beta_0} = \frac{\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)}{\sigma^2} = 0 \Rightarrow \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\frac{\partial l}{\partial \beta_1} = \frac{\sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i)}{\sigma^2} = 0 \Rightarrow \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$$

β_0 和 β_1 的估计量

where $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n(\bar{x})^2$,

$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}$

Proof

$$\sum_{i=1}^n x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\Rightarrow \sum_{i=1}^n x_i y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) \sum_{i=1}^n x_i - \hat{\beta}_1 \sum_{i=1}^n x_i^2 = 0$$

$$\Rightarrow \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y} + (n(\bar{x})^2 - \sum_{i=1}^n x_i^2) \hat{\beta}_1 = 0$$

$$\Rightarrow \hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n(\bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

中心化
向量 $\vec{x} = \begin{pmatrix} x_1 - \bar{x} \\ \vdots \\ x_n - \bar{x} \end{pmatrix}$

向量 $\vec{y} = \begin{pmatrix} y_1 - \bar{y} \\ \vdots \\ y_n - \bar{y} \end{pmatrix}$

内积

$$S_{xx} = \vec{x}^T \vec{x} = \vec{x} \cdot \vec{x}$$

$$S_{xy} = \vec{x} \cdot \vec{y} = \vec{x}^T \vec{y}$$

残差平方和估计量

Estimation of σ^2

Residuals

$\hat{y}_1 - \hat{y}_2 = \hat{\varepsilon}_i = (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i), \sum_{i=1}^n \hat{\varepsilon}_i = 0$

The residual sum of square (error sum of square):

$SSE = \sum_{i=1}^n \hat{\varepsilon}_i^2 = S_{yy} - \frac{S_{xy}^2}{S_{xx}}$

where $S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n y_i^2 - n(\bar{y})^2$

Proof

$$\begin{aligned} \sum_{i=1}^n \hat{\varepsilon}_i^2 &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = \sum_{i=1}^n (y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) - \hat{\beta}_1 x_i)^2 \\ &= \sum_{i=1}^n ((y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x}))^2 \\ &= \sum_{i=1}^n (y_i - \bar{y})^2 + \hat{\beta}_1^2 \sum_{i=1}^n (x_i - \bar{x})^2 - 2 \hat{\beta}_1 \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \\ &= S_{yy} + \left(\frac{S_{xy}}{S_{xx}}\right)^2 S_{xx} - 2 \frac{S_{xy}}{S_{xx}} S_{xy} = S_{yy} - \frac{S_{xy}^2}{S_{xx}} \end{aligned}$$

Estimate of the error variance σ^2 :

$\hat{\sigma}^2 = MSE = \frac{SSE}{n-2}$, where n is the number of observations



$\sum \hat{\varepsilon}_i = 0$

$\frac{1}{n} \sum_{i=1}^n \hat{\varepsilon}_i = 0$

残差平方和

中, S_{xy} 是向量的内积

$\frac{(n-2) \hat{\sigma}^2}{\sigma^2} \sim \chi_{n-2}^2$

Example 11.1 (continued)

	x	y	x^2	y^2	xy	$\hat{y}(= \hat{\beta}_0 + \hat{\beta}_1 x)$	$\hat{\varepsilon}$
	3	9	9	81	27	7.15	1.85
	3	5	9	25	15	7.15	-2.15
	4	12	16	144	48	9.89	2.11
	5	9	25	81	45	12.63	-3.63
	6	14	36	196	84	15.37	-1.37
	6	16	36	256	96	15.37	0.63
	7	22	49	484	154	18.11	3.89
	8	18	64	324	144	20.85	-2.85
	8	24	64	576	192	20.85	3.15
	9	22	81	484	198	23.59	-1.59
Total	59	151	389	2651	1003		0.04 (rounding error)

Example 11.1.1 (continued)

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$\bar{x} = 5.9, \quad \bar{y} = 15.1$$

$$\hat{\varepsilon}_i = (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i), \quad \sum_{i=1}^n \hat{\varepsilon}_i = 0$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$$

$$S_{xx} = \sum_{i=1}^n x_i^2 - n(\bar{x})^2 = 389 - 10(5.9)^2 = 40.9$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$S_{xy} = \sum_{i=1}^n x_i y_i - n \bar{x} \bar{y} = 1003 - 10(5.9)(15.1) = 112.1$$

$$S_{yy} = \sum_{i=1}^n y_i^2 - n(\bar{y})^2 = 2651 - 10(15.1)^2 = 370.9$$

$$r^2 = \frac{SSE}{n-2}$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{112.1}{40.9} = 2.74$$

$$SSE = \sum_{i=1}^n (\hat{\varepsilon}_i)^2$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 15.1 - (2.74)(5.9) = -1.07$$

$$SSE = S_{yy} - \frac{(S_{xy})^2}{S_{xx}} = 370.9 - \frac{(112.1)^2}{40.9} = 63.653$$

$$= \sum y_i^2 - \frac{(\sum x_i y_i)^2}{\sum x_i^2}$$

Example 11.1.1 (continued)

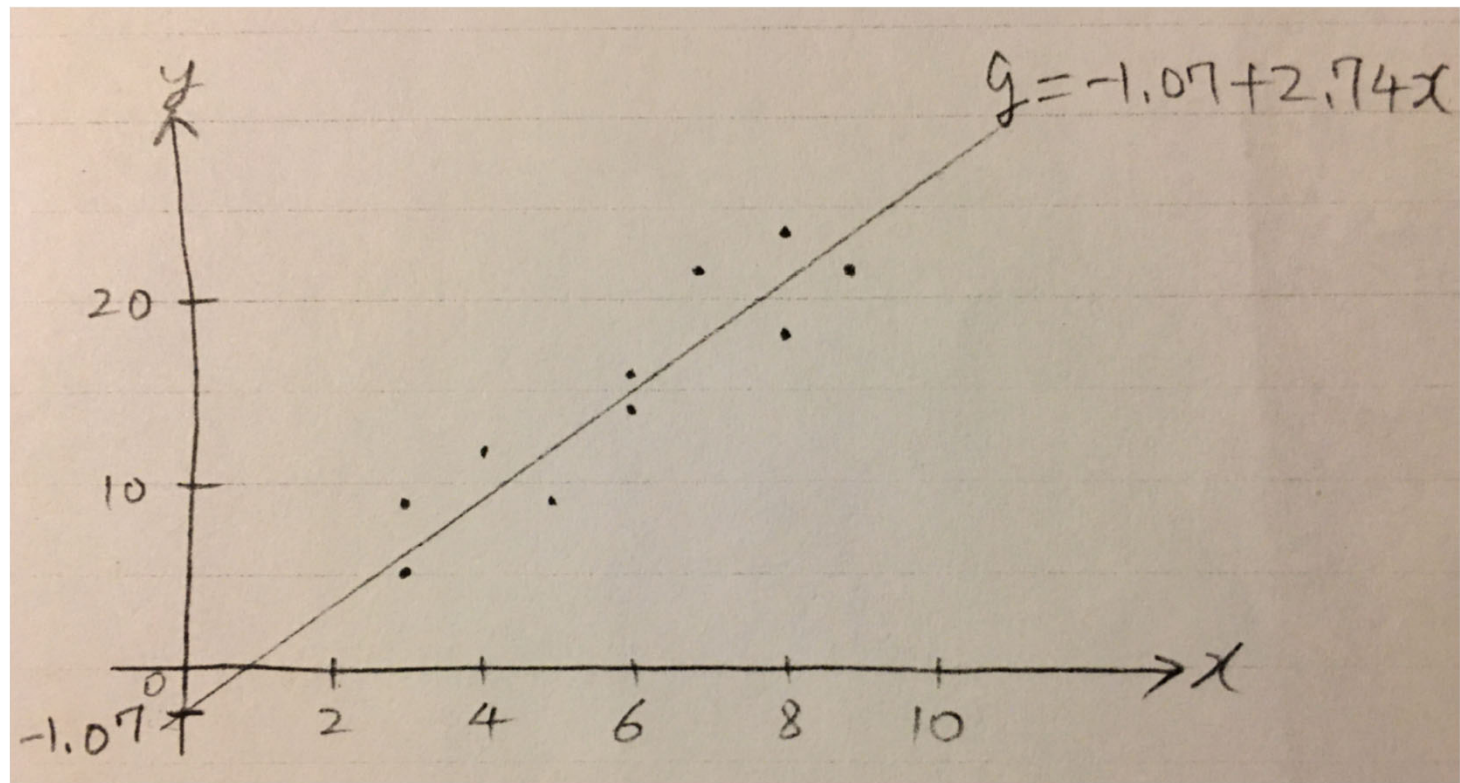
$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x = -1.07 + 2.74x$$

$$\hat{\sigma}^2 = \frac{SSE}{n-2} = \frac{63.653}{8} = 7.96$$

经带回归直线(方程)

误差的无偏估计

The mean of the duration of relief increases by 2.74 if the dosage increases by 1.



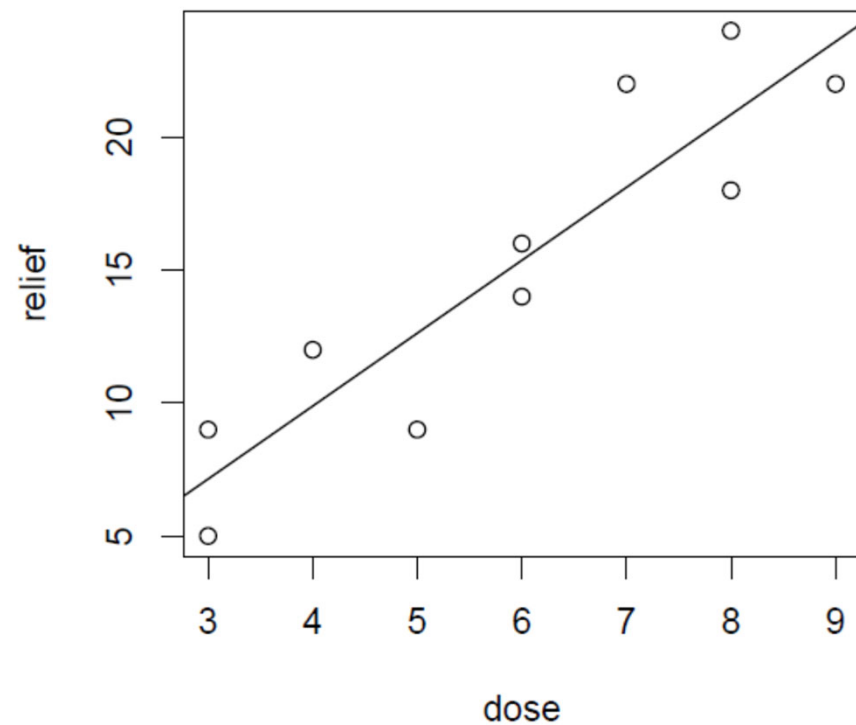
Regression fit in R

$x =$
 $y =$

```
dose=c(3,3,4,5,6,6,7,8,8,9)
relief=c(9,5,12,9,14,16,22,18,24,22)
fit=lm(relief ~ dose)
# or fit=lm(relief~dose, data=dataset)
summary(fit)
fit$coefficients
summary(fit)$sigma^2
plot(dose,relief)
abline(lm(relief ~ dose)))
```

$fit = lm(y \sim x)$

summary(fit)



Output in R

```
> summary(fit)
```

Call:

```
lm(formula = relief ~ dose)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.6333	-2.0128	-0.3741	2.0428	3.8851

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.0709	2.7509	-0.389	0.707219
dose	2.7408	0.4411	6.214	0.000255 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.821 on 8 degrees of freedom

Multiple R-squared: 0.8284, Adjusted R-squared: 0.8069

F-statistic: 38.62 on 1 and 8 DF, p-value: 0.0002555

调整R平方

$$= 1 - \frac{\text{残差平方和} / \text{自由度}}{\text{总平方和} / \text{自由度}}$$

调整R平方 = $\frac{\text{回归平方和}}{\text{总平方和}} = 1 - \frac{\text{残差平方和}}{\text{总平方和}}$

Calculation by yourself in R

```
dose=c(3,3,4,5,6,6,7,8,8,9)
relief=c(9,5,12,9,14,16,22,18,24,22)
n=10
S_yy = (n-1) * var(relief)
S_xx = (n-1) * var(dose)
S_xy = (n-1) * cov(dose,relief)
b1=S_xy/S_xx
x_bar=mean(dose)
y_bar=mean(relief)
b0=y_bar-b1*x_bar
SSE=sum(fit$residuals^2)
sigma2_hat=SSE/(n-2)
```



11.3 Inferences about Regression Parameters

Distribution of $\hat{\beta}_1$

$$E(Y_i) = \beta_0 + \beta_1 x_i$$

$$E(\hat{\beta}_1) = E\left(\frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2}\right) = \frac{\sum_{i=1}^n (x_i - \bar{x})E(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$= \frac{\beta_1 \sum_{i=1}^n (x_i - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2} = \beta_1$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})y_i - \bar{y} \sum_{i=1}^n (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$= \frac{\sum_{i=1}^n (x_i - \bar{x})y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\text{Var}(\hat{\beta}_1) = \text{Var}\left(\frac{\sum_{i=1}^n (x_i - \bar{x})Y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}\right) = \frac{\sum_{i=1}^n (x_i - \bar{x})^2 \text{Var}(Y_i)}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})^2 \sigma^2}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2}$$

$$= \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sigma^2}{S_{xx}}$$

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}\right)$$

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$\bar{y} = \beta_0 + \beta_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n y_i = \frac{1}{n} \sum_{i=1}^n (\beta_0 + \beta_1 x_i + \varepsilon_i)$$

Independence of $\bar{Y}$ and $\hat{\beta}_1$

$\text{Var}(\bar{Y})$

$$\mathbb{E} \bar{Y} = \beta_0 + \beta_1 \bar{x}$$

Recall

$$= \text{Var}(\bar{\varepsilon})$$

$$\bar{Y} \sim N\left(\beta_0 + \beta_1 \bar{x}, \frac{\sigma^2}{n}\right) \text{ and } \hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}\right).$$

$$= \frac{\sigma^2}{n}$$

$$\text{Let } w_i = \frac{x_i - \bar{x}}{\sum_{i=1}^n (x_i - \bar{x})^2}. \text{ Then } \hat{\beta}_1 = \sum_{i=1}^n w_i y_i.$$

$$\bar{\varepsilon} = \frac{1}{n} \sum_{i=1}^n \varepsilon_i$$

$$\text{Cov}(\bar{Y}, \hat{\beta}_1) = \text{Cov}\left(\frac{1}{n} \sum_{i=1}^n Y_i, \sum_{i=1}^n w_i Y_i\right)$$

$$= \frac{1}{n} \sum_{i=1}^n \text{Cov}(Y_i, w_i Y_i) + \frac{1}{n} \sum_{i \neq j} \text{Cov}(Y_i, w_j Y_j) = \frac{1}{n} \sum_{i=1}^n w_i \text{Var}(Y_i)$$

$$= \frac{1}{n} \sum_{i=1}^n w_i \sigma^2 = \frac{\sigma^2}{n} \sum_{i=1}^n w_i = 0$$

Therefore, $\bar{Y}$ and $\hat{\beta}_1$ are independent.

Distribution of $\hat{\beta}_0$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$$

$$\bar{Y} \sim N\left(\beta_0 + \beta_1 \bar{x}, \frac{\sigma^2}{n}\right) \text{ and } \hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}\right)$$

$$E(\hat{\beta}_0) = E(\bar{Y} - \hat{\beta}_1 \bar{x}) = E(\bar{Y}) - E(\hat{\beta}_1) \bar{x} = \beta_0 + \beta_1 \bar{x} - \beta_1 \bar{x} = \beta_0$$

$$\begin{aligned} \text{Var}(\hat{\beta}_0) &= \text{Var}(\bar{Y} - \hat{\beta}_1 \bar{x}) = \text{Var}(\bar{Y}) + \bar{x}^2 \text{Var}(\hat{\beta}_1) \\ &= \frac{\sigma^2}{n} + \frac{\bar{x}^2 \sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} = \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) \sigma^2 \end{aligned}$$

$$\hat{\beta}_0 \sim N\left(\beta_0, \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right) \sigma^2\right)$$

Inference about $\hat{\beta}_1$ and $\hat{\beta}_0$: based on the t distribution

Inference about $\hat{\beta}_1$

$$T = \frac{\hat{\beta}_1 - \beta_1}{S / \sqrt{S_{xx}}},$$

$$\text{df} = n - 2$$

$$\frac{\hat{\beta}_1 - \beta_1}{S / \sqrt{S_{xx}}} \sim N(0, 1)$$

Inference about $\hat{\beta}_0$

$$T = \frac{\hat{\beta}_0 - \beta_0}{S \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}},$$

$$\text{df} = n - 2$$

$$\frac{\hat{\beta}_0 - \beta_0}{S \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}} \sim N(0, 1)$$

where $S = \sqrt{\frac{\text{SSE}}{n-2}} = \sqrt{\hat{\sigma}^2}$

但 σ^2 未知

故用 $\hat{\sigma}^2$ 估计

而 $\frac{(n-2) \hat{\sigma}^2}{n-2} \sim \chi^2_{n-2}$, 且 $\hat{\sigma}^2$ 与 $\hat{\beta}_1$ 和 $\hat{\beta}_0$ 独立

A $100(1-\alpha)\%$ CI for β_1 :

$$\left(\hat{\beta}_1 - t_{\alpha/2, n-2} \frac{s}{\sqrt{s_{xx}}}, \hat{\beta}_1 + t_{\alpha/2, n-2} \frac{s}{\sqrt{s_{xx}}} \right)$$

Example 11.1.1 (continued) Construct a 95% CI for the slope of the regression line.

Dosage (x)	3	3	4	5	6	6	7	8	8	9
Duration of relief (y)	9	5	12	9	14	16	22	18	24	22

$$\hat{\beta}_1 \pm t_{0.025, 8} \frac{s}{\sqrt{s_{xx}}} = 2.74 \pm 2.306 \frac{2.821}{\sqrt{40.9}} = 2.74 \pm 1.02$$

The 95% CI is (1.72, 3.76).

$0 \notin (1.72, 3.76) \Rightarrow \beta_1$ is significantly different from 0 at $\alpha = 0.05$.

A $100(1-\alpha)\%$ CI for β_0 :

$$\left(\hat{\beta}_0 - t_{\alpha/2, n-2} s \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}}}, \hat{\beta}_0 + t_{\alpha/2, n-2} s \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}}} \right)$$

Example 11.1.1 (continued) Construct a 95% CI for β_0 .

$$\begin{aligned} \hat{\beta}_0 \pm t_{0.025, 8} s \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{s_{xx}}} &= -1.07 \pm 2.306 \cdot 2.821 \sqrt{\frac{1}{10} + \frac{(5.9)^2}{40.9}} \\ &= -1.07 \pm 6.34 \end{aligned}$$

The 95% CI is $(-7.41, 5.27)$.

$0 \in (-7.41, 5.27) \Rightarrow \beta_0$ is not significantly different from 0 at $\alpha = 0.05$.

Testing hypothesis for β_1

	Case (a)	Case (b)	Case (c)
Step 1	$H_0: \beta_1 = 0, H_1: \beta_1 \neq 0$ (2-sided alternative)	$H_0: \beta_1 = 0, H_1: \beta_1 > 0$ (1-sided alternative)	$H_0: \beta_1 = 0, H_1: \beta_1 < 0$ (1-sided alternative)
Step 2	$\alpha = ?$	$\alpha = ?$	$\alpha = ?$
Step 3	$t = \frac{\hat{\beta}_1}{s/\sqrt{s_{xx}}}$ Rejection Region: $t \geq t_{\alpha/2, n-2}$ <i>(Handwritten: $s(\hat{\beta}_1)$)</i>	$t = \frac{\hat{\beta}_1}{s/\sqrt{s_{xx}}}$ Rejection Region: $t \geq t_{\alpha, n-2}$ <i>(Handwritten: $s = \sqrt{\frac{SSE}{n-2}}$)</i>	$t = \frac{\hat{\beta}_1}{s/\sqrt{s_{xx}}}$ Rejection Region: $t \leq -t_{\alpha, n-2}$
Step 4	$t?$ Decision <i>(Handwritten: $T \sim t_{n-2}$)</i>	$t?$ Decision	$t?$ Decision
Step 5	$p = 2P(T \geq t)$	$p = P(T \geq t)$	$p = P(T \leq t)$

Example 11.1.1 (continued) Do the data constitute strong evidence that the mean duration of relief increases with dosages of the drug? Test using $\alpha = 0.05$.

$$H_0: \beta_1 = 0, H_1: \beta_1 > 0$$

$$\alpha = 0.05$$

$$\text{Test statistic: } T = \frac{\hat{\beta}_1}{S/\sqrt{S_{xx}}}$$

$$\text{Rejection region: } t \geq t_{\alpha, n-2} = t_{0.05, 8} = 1.860$$

$$t = \frac{2.74}{2.821/\sqrt{40.9}} = 6.213 > 1.860 \Rightarrow \text{Reject } H_0$$

$$p\text{-value} = \underline{P(T > 6.213)} = 0.00013$$

Yes, the data suggest that the mean duration of relieve significantly increases with dosages at level $\alpha = 0.05$.

In R,

$$1\text{-pt}(6.213, 8)$$

$$= 1 - P(T \leq 6.123)$$

CI and Testing for the parameters in R

```
dose=c(3,3,4,5,6,6,7,8,8,9)
relief=c(9,5,12,9,14,16,22,18,24,22)
fit=lm(relief ~ dose)
## 95% & 99% confidence intervals for the parameters, respectively
confint(fit)
confint(fit,level=0.99)
## Testing for the parameters
summary(fit)
### Calculation by yourself #####
b0=fit$coefficients[1]
b1=fit$coefficients[2]
n=10
## 95% confidence Interval of the slope
E= qt(0.975,n-2)*summary(fit)$coefficients[2,2]
CI95b1=c(b1-E,b1+E)
## 95% confidence Interval of the intercept
E= qt(0.975,n-2)*summary(fit)$coefficients[2,1]
CI95b0=c(b0-E,b0+E)
```



11.4 Predicting New y-values Using Regression

At a specified value $x = x^*$, the expected value of y is $y^* = \beta_0 + \beta_1 x^*$ and this is estimated by $\hat{\beta}_0 + \hat{\beta}_1 x^*$. Estimated standard error of y^* is

$$\text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x^*) = \text{Var}(\hat{\beta}_0) + (x^*)^2 \text{Var}(\hat{\beta}_1) + 2 \text{Cov}(\hat{\beta}_0, \hat{\beta}_1 x^*)$$

$$S \sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}}$$

Inferences about $\beta_0 + \beta_1 x^*$: based on the t distribution

$$\text{Cov}(\bar{y} - \hat{\beta}_1 \bar{x}, \hat{\beta}_1 x^*)$$

$$= x^* \text{Cov}(\bar{y}, \hat{\beta}_1)$$

$$= -\bar{x} x^* \text{Cov}(\hat{\beta}_0, \hat{\beta}_1)$$

$$= 0 - \bar{x} x^* \text{Var}(\hat{\beta}_1)$$

$$\text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x^*)$$

$$= \text{Var}(\hat{\beta}_0) + (x^*)^2 \text{Var}(\hat{\beta}_1)$$

$$+ 2 \bar{x} x^* \text{Cov}(\hat{\beta}_0, \hat{\beta}_1)$$

$$T = \frac{(\hat{\beta}_0 + \hat{\beta}_1 x^*) - (\beta_0 + \beta_1 x^*)}{S \sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}}}, \quad \text{df} = n - 2$$

$$= \left(\frac{1}{n} + \frac{(\bar{x})^2}{S_{xx}} \right) \sigma^2 + (x^*)^2 \frac{1}{S_{xx}} \sigma^2$$

$$- 2 \bar{x} x^* \frac{1}{S_{xx}} \sigma^2$$

$$= \left[\frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}} \right] \sigma^2$$

$$\sigma^2 \left(\frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}} \right)$$

it

A 100(1- α)% CI for the expected response $y^* = \beta_0 + \beta_1 x^*$:

$$\hat{\beta}_0 + \hat{\beta}_1 x^* \pm t_{\alpha/2, n-2} s \sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}}$$

Example 11.1.1 (continued) Obtain a 99% CI for the expected duration of relief when the dosage is (a) $x^* = 6$ and (b) $x^* = 9.5$.

(a) $\hat{\beta}_0 + \hat{\beta}_1 x^* = -1.07 + 2.74 \cdot 6 = 15.37$

$$\begin{aligned} 99\% \text{ CI: } 15.37 \pm t_{\alpha/2, n-2} 2.821 \sqrt{\frac{1}{10} + \frac{(6-5.9)^2}{40.9}} &= 15.37 \pm 3.355 \cdot 0.893 \\ &= 15.37 \pm 2.996 \end{aligned}$$

or (12.374, 18.366)

(b) $\hat{\beta}_0 + \hat{\beta}_1 x^* = -1.07 + 2.74 \cdot 9.5 = 24.96$

$$99\% \text{ CI: } 24.96 \pm 3.355 \cdot 2.821 \sqrt{\frac{1}{10} + \frac{(9.5-5.9)^2}{40.9}} = 24.96 \pm 6.11$$

or (18.85, 31.07)

Prediction of a single response for a specified x value



The estimated s.e. when predicting a single observation y at a given x^* is

$$s \sqrt{1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}}.$$

A $100(1-\alpha)\%$ prediction interval for a future trial:

$$\hat{\beta}_0 + \hat{\beta}_1 x^* \pm t_{\alpha/2, n-2} s \sqrt{1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}}$$

Example 11.1.1 (continued) A new trial is to be made on a single patient with the dosage $x^* = 6.5$ mg. Predict the duration of relief and give a 95% prediction interval.

$$\begin{aligned} & \hat{\beta}_0 + \hat{\beta}_1 x^* \pm t_{0.025,8} s \sqrt{1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}} \\ &= -1.07 + 2.74 \cdot 6.5 \pm 2.306 \cdot 2.821 \sqrt{1 + \frac{1}{10} + \frac{(6.5 - 5.9)^2}{40.9}} \\ &= 16.74 \pm 6.85 \\ &\text{or } (9.89, 23.59) \end{aligned}$$

Confidence Interval and Prediction Interval in R

```
dose=c(3,3,4,5,6,6,7,8,8,9)
relief=c(9,5,12,9,14,16,22,18,24,22)
fit=lm(relief ~ dose)
```



99% confidence intervals for x=6 and x=9.5, respectively

```
new=data.frame(dose=6)
predict(fit, new, interval="confidence", level=0.99)
new=data.frame(dose=9.5)
predict(fit, new, interval="confidence", level=0.99)
```

95% prediction intervals for x=6.5

```
new=data.frame(dose=6.5)
predict(fit, new, interval="prediction", level=0.95)
```

Example 11.3.1 In a study to determine how the skill in doing a complex assembly job is influenced by the amount of training, 15 new recruits were given varying amounts of training ranging between 3 and 12 hours. After the training, their times to perform the job were recorded. Denote x = duration of training (in hours), and y = time to do the job (in minutes). The following were calculated: $\bar{x} = 7.2$, $\bar{y} = 45.6$, $s_{xx} = 33.6$, $s_{yy} = 160.2$, $s_{xy} = -57.2$

(a) Determine the equation of the best fitting straight line.

$$\hat{\beta}_1 = \frac{s_{xy}}{s_{xx}} = \frac{-57.2}{33.6} = -1.702$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 45.6 - (-1.702)(7.2) = 57.85$$

$$\therefore \hat{y} = 57.85 - 1.702x$$

Example 11.3.1 (continued)

(b) Do the data substantiate the claim that the job time decreases with more hours of training? Test using $\alpha = 0.01$.

$$H_0: \beta_1 = 0, H_1: \beta_1 < 0$$

$$\alpha = 0.01$$

$$\text{Test statistic: } T = \frac{\hat{\beta}_1}{S/\sqrt{S_{xx}}}$$

$$\text{Rejection region: } t < -t_{\alpha, n-2} = -t_{0.01, 13} = -2.650$$

$$SSE = S_{yy} - \frac{S_{xy}^2}{S_{xx}} = 160.2 - \frac{(-57.2)^2}{33.6} = 62.824$$

$$\hat{\sigma}^2 = \frac{SSE}{n-2} = \frac{62.824}{13} = 4.833$$

$$t = \frac{-1.702}{\sqrt{4.833/33.6}} = -4.49 < -2.650 \Rightarrow \text{Reject } H_0$$

$$p\text{-value} = P(T < -4.49) = 0.0003$$

Yes, the data suggest that the job time decreases with more hours of training at level $\alpha = 0.05$.

In R,

`pt(-4.49, 13)`

Example 11.3.1 (continued)

(c) Estimate the mean job time for 9 hours of training and construct a 95% CI.

The expected job time corresponding to $x^* = 9$ hours is

$$\hat{\beta}_0 + \hat{\beta}_1 x^* = 57.85 + (-1.702) \cdot 9 = 42.53 \text{ minutes}$$

The 95% CI is

$$\begin{aligned} & \hat{\beta}_0 + \hat{\beta}_1 x^* \pm t_{0.025,13} s \sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}} \\ &= 42.53 \pm 2.160 \sqrt{4.833} \sqrt{\frac{1}{15} + \frac{(9 - 7.2)^2}{33.6}} = 42.53 \pm 1.92 \\ & \text{or } (40.61, 44.45). \end{aligned}$$

Example 11.3.1 (continued)

(d) Find the predicted y for $x = 35$ hours and comment on the result.

The predicted job time is

$$\hat{\beta}_0 + \hat{\beta}_1 x^* = 57.85 + (-1.702) \cdot 35 = -1.72 \text{ minutes}$$

It is not reasonable.

11.5 Estimating Lack of Fit in Linear Regression

Model assumptions

$$Y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2), \quad i = 1, \dots, n$$

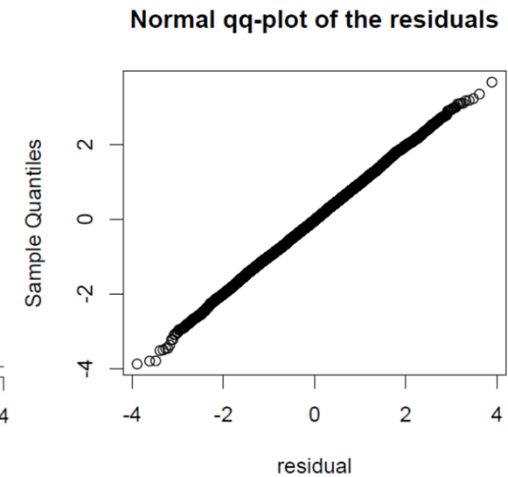
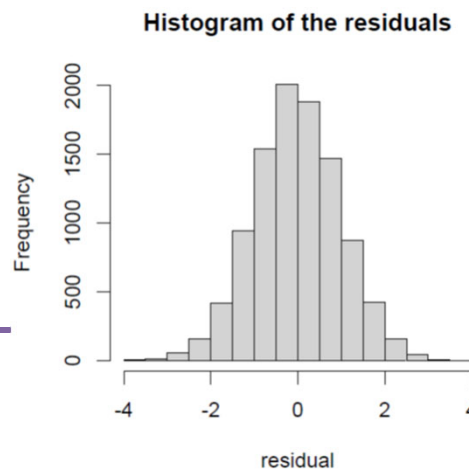
1. **Linearity:** The underlying regression is linear.
2. **Independence:** The errors are independent.
3. **Homoscedasticity:** The variances are constant.
4. **Normality:** Each observation/error follows a normal distribution.

回归诊断

Diagnostic Test

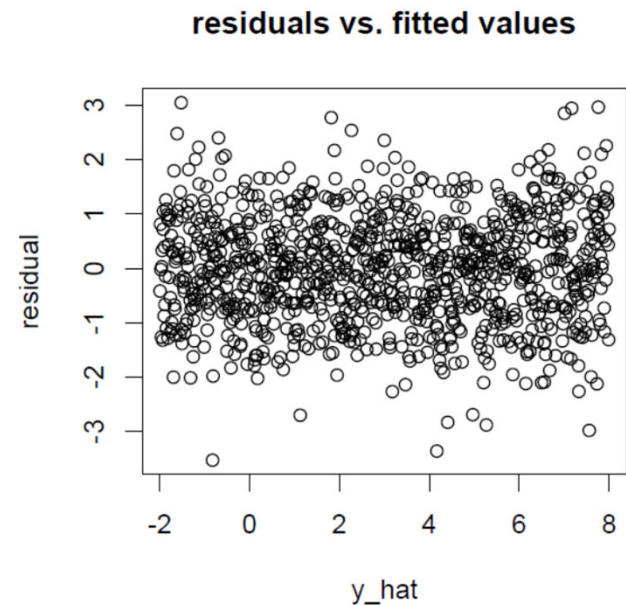
(4) Normality

- Histogram or dot diagram of residuals
- Plot of residual vs. normal (normal qq-plot)



(3) Heteroscedasticity

- Plot of residual vs. $\hat{y}$



The examination of residuals

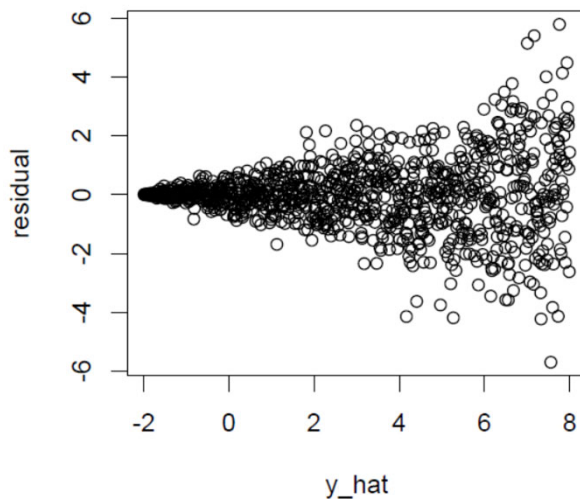
(A) Violation of homoscedasticity (3): Variance is not constant.

Need a weighted least squares or a transformation on y (eg. $\log y$)

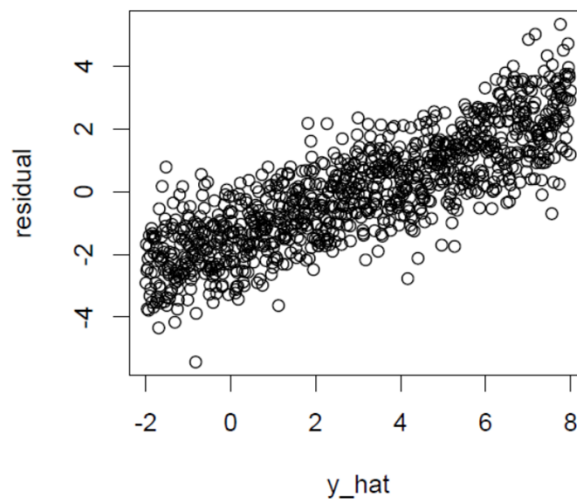
(B) Error in analysis: The effect can also be caused by erroneously omitting a β_0 term in the model.

(C) Inadequate model: Violation of (1). Need for extra term(s) in the model (eg., square or cross-product terms) or a transformation on y .

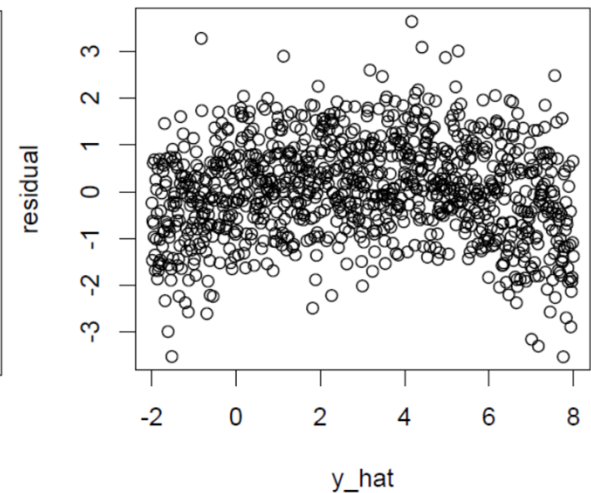
(A) Heteroscedasticity



(B) Error in regression analysis



(C) Inadequate model



11.6 Correlation

$$y_i = (\hat{\beta}_0 + \hat{\beta}_1 x_i) + (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)$$

Observed y value	Explained by linear relation	Residual or deviation from linear relation
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$$SSE = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = S_{yy} - \frac{S_{xy}^2}{S_{xx}}$$

Decomposition of variability

S_{yy}	$= \frac{S_{xy}^2}{S_{xx}} +$	SSE
(SST)	(SSR)	
Total variability of y	Variability explained by the linear regression	residual or unexplained variability

$$\frac{SSR}{SST} = \frac{SST - SSE}{SST} = \frac{S_{xy}^2 / S_{xx}}{S_{yy}} = \frac{S_{xy}^2}{S_{xx} S_{yy}} = R^2$$

$R^2 = \frac{S_{xy}^2}{S_{xx}S_{yy}}$: percentage of the variation in y which can be explained by the linear regression

$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$: sample correlation coefficient

Example 11.1.1 (continued)

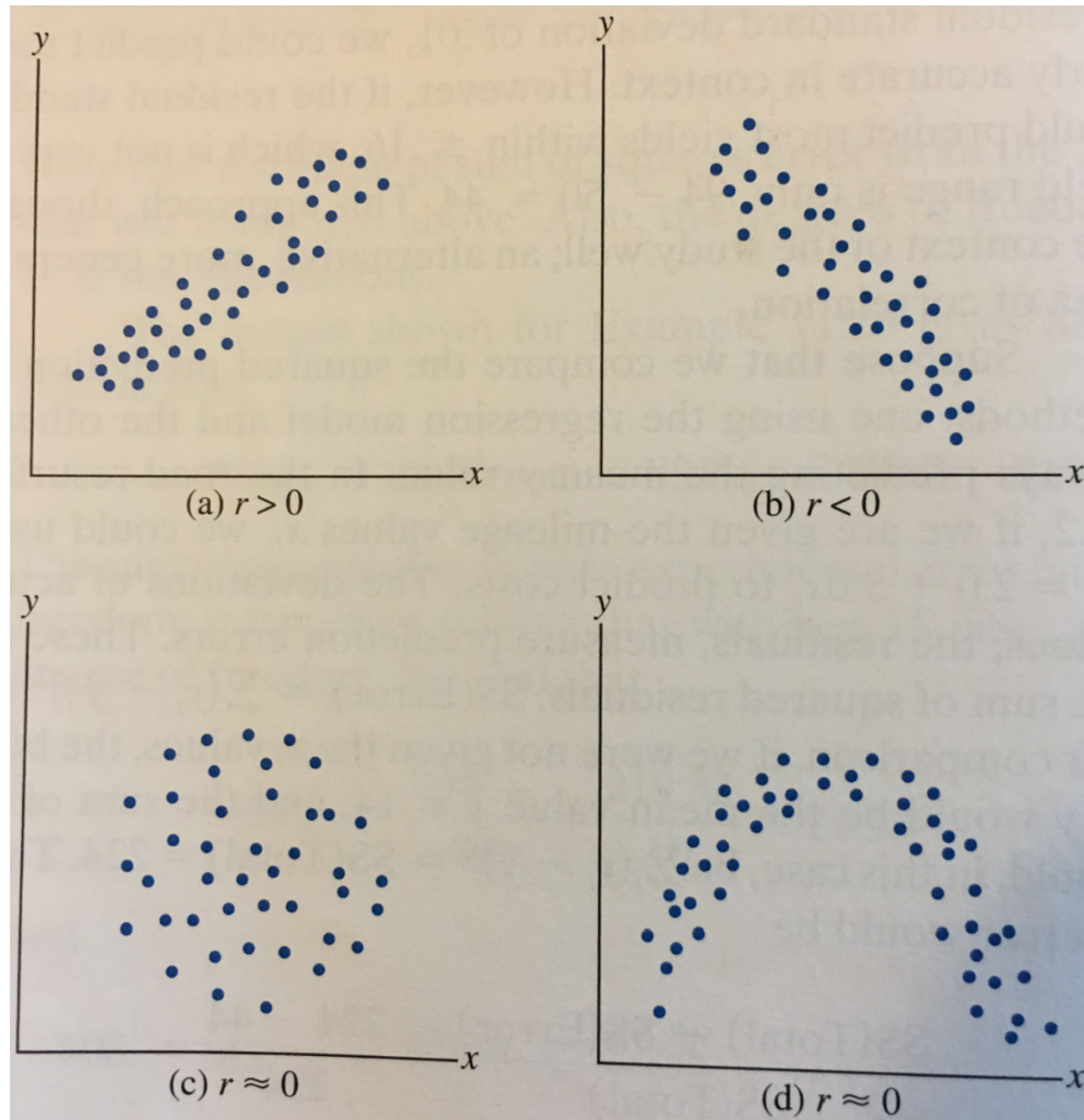
$$S_{xx} = 40.9, S_{yy} = 370.9, S_{xy} = 112.1$$

$$R^2 = \frac{S_{xy}^2}{S_{xx}S_{yy}} = \frac{(112.1)^2}{40.9 \cdot 370.9} = 0.83$$

83% of the variability in y is explained by linear regression.

$$r = \sqrt{0.83} = 0.91$$

Correlation coefficient



ANOVA Table

Degrees of freedom:

SSE has df $n - 2$, SST has df $n - 1$.

$$\begin{aligned} SST &= \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (y_i - \hat{y}_i + \hat{y}_i - \bar{y})^2 \\ &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + 2 \sum_{i=1}^n (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) \\ &= SSE + SSR + 0 \end{aligned}$$

SSR has df 1.

$$\frac{SST}{\sigma^2} \sim \chi^2_{n-1}$$

$$\frac{SSR}{\sigma^2} \sim \chi^2_1$$

$$\frac{SSE}{\sigma^2} \sim \chi^2_{n-2}$$

Source	SS	DF	MS	F
Regression	$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$	1	$MSR = \frac{SSR}{1}$	$F = \frac{MSR}{MSE}$
Error	$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$	$n - 2$	$MSE = \frac{SSE}{n-2} = \hat{\sigma}^2$	
Total	$SST = \sum_{i=1}^n (y_i - \bar{y})^2$	$n - 1$		

$$\sim F_{1, n-2}$$

Example 11.1.1 (continued)

$$SSE = 63.65$$

$$SST = S_{yy} = 370.9$$

$$SSR = SST - SSE = 370.9 - 63.65 = 307.25$$

Source	SS	DF	MS	F
Regression	$SSR = 307.25$	1	$MSR = \frac{307.25}{1} = 307.25$	$F = \frac{307.25}{7.96} = 38.6$
Error	$SSE = 63.65$	$n - 2$	$MSE = \frac{63.65}{8} = 7.96$	
Total	$SST = 370.9$	$n - 1$		

ANOVA in R

```
dose=c(3,3,4,5,6,6,7,8,8,9)
relief=c(9,5,12,9,14,16,22,18,24,22)
fit=lm(relief ~ dose)
anova(fit)
## R square
cov(dose, relief)
cor(dose,relief)
```

